Lab8: Jobs Control

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1. 題目要求截圖

   1. 透過 pstree 或 jobs 展示有 3 個 vim 在背景執行

```
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ vim a.txt &
vim b.txt &
vim c.txt &
[1] 2905
[2] 2906
[3] 2907
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ jobs
[1]   Stopped                 vim a.txt
[2]+  Stopped                 vim b.txt
[3]-  Stopped                 vim c.txt
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ fg %1
vim a.txt
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ jobs
[2]+  Stopped                 vim b.txt
[3]   Stopped                 vim c.txt
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ echo $$
2402
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ pstree -p 2402
bash(2402)─┬─pstree(2915)
           ├─vim(2906)
           └─vim(2907)
```

   2. 使用 kill 來結束 vim b.txt
   3. 使用 fg 來結束 vim c.txt

```
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ kill 2906
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ jobs
[2]+  Stopped                 vim b.txt
[3]   Stopped                 vim c.txt
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ pstree -p 2402
bash(2402)─┬─pstree(2919)
           ├─vim(2906)
           └─vim(2907)
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ kill -9 2906
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ jobs
pstree -p 2402
[2]+  Killed                  vim b.txt
[3]   Stopped                 vim c.txt
bash(2402)─┬─pstree(2926)
           └─vim(2907)
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ fg %3
vim c.txt
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ jobs
pstree -p 2402
bash(2402)──pstree(2929)
```

4. 使用 renice 將 vim d.txt 的 priority + 10
5. 透過 ps-l 來展示更新之後的 job priority

```
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ vim d.txt &
[1] 2932
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ ps -l
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1000    2402    2376  0  80   0 -  3073 do_wai pts/0    00:00:00 bash
0 T  1000    2932    2402  0  80   0 - 16203 do_sig pts/0    00:00:00 vim
0 R  1000    2934    2402  0  80   0 -  3325 -      pts/0    00:00:00 ps

[1]+  Stopped                 vim d.txt
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ renice +10 -p 2932
2932 (process ID) old priority 0, new priority 10
f74136047@f74136047-VirtualBox:~/Linux2025Fall/Lab8_Jobs$ ps -l
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1000    2402    2376  0  80   0 -  3073 do_wai pts/0    00:00:00 bash
0 T  1000    2932    2402  0  90  10 - 16203 do_sig pts/0    00:00:00 vim
0 R  1000    2939    2402  0  80   0 -  3325 -      pts/0    00:00:00 ps
```

2. 本次作業/課程心得

這次作業遇到一個小問題，是原先輸入 kill 2906，但 vim b.txt 沒有如預期結束運作。

後來發現，因為 vim b.txt 是「Stopped」狀態，單純的 kill 預設是 SIGTERM，不會立刻終止暫停中的程序。

需要改用更強的訊號: kill -9 2906，送出 SIGKILL，強制結束該行程。

嘗試後果然 vim b.txt 確實被殺掉，只剩下 vim c.txt (2907)。